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Literature Review

This paper: (i) full commitment; (ii) random changes in buyers' values; (iii) infinite horizon with continuous time; (iv) a continuum of buyers

- Randomness in values: Conlisk (1984); Biehl (2001); Deb (2014)
  - None find repeated discounting except this paper.
- No commitment: Conlisk, Gerstner & Sobel (1984); Sobel (1991)
  - With (i) infinite horizon, (ii) finite buyers entering at each period and (iii) binary buyer types, the monopolist's strategy in each stationary equilibrium features *price cycles*.
- Beyond durable goods: Varian (1980); Maskin & Tirole (1988)
  - Varian: Informed and uninformed buyers (search frictions). A static setting in which sellers set prices probabilistically.
  - Maskin & Tirole: No micro foundation of buyers. No commitment. Multiple MPE include the *kinked demand curve* equilibria and the *Edgeworth cycles*.

Buyers and Payoffs

Time: infinite horizon, continuous

Buyers: infinitesimal, each with unit demand

- Initial mass I
- Constant arrival rate λ
- Exogenous shock rate $\rho > 0 \Rightarrow$ leave forever w/o purchases
- Changing values at t: $\theta_t \in \{\theta_L, \theta_H\}$
- Common discount rate r

Assume for stationarity $I = \lambda/\rho$.

Processes for Values

Upon arrival date τ, buyers have θ_H with prob. γ.

A homogeneous continuous-time Markov process:

- Low-to-high rate α_L; High-to-low rate α_H
- Assume for stationarity $\gamma = \gamma^S = \dfrac{\alpha_L}{\alpha_L + \alpha_H}$.

The stock at any time: $I\gamma^S$ high valuers, $I(1-\gamma^S)$ low valuers

Players' Strategies

Seller commits at $t = 0$ to a price path $p : \mathbb{R}_+ \to \mathbb{R}_+$.

- Deterministic paths are WLOG (Strausz, 2006).

A buyer's problem of purchase date: an optimal stopping problem

- Let Σ be the set of Markov strategies.
  - Functions $\sigma(\theta_L, \cdot), \sigma(\theta_H, \cdot) : \mathbb{R}_+ \to \{0, 1\}$, right continuous
  - WLOG given that the value process is Markov
- For any arrival date τ and $\theta^{[\tau,\infty)} := \{\theta_s : s \geq \tau\}$, a buyer's purchase date when playing $\sigma \in \Sigma$:

$$\mu_\sigma(\theta^{[\tau,\infty)}) = \inf\{s \geq \tau : \sigma(\theta_s, s) = 1\},$$

provided he has not left.

Price-Path Mechanisms

A price-path mechanism $\mathcal{M}_P = \langle p, x \rangle$ includes the prescription $x \in \Sigma$.

A buyer is to purchase the good at t paying $p(t)$ if and only if

- He has arrived to the market by t and has not left.
- He has not yet purchased.
- His value at t is θ_t, and $x(\theta_t, t) = 1$.

Buyer's Problem

A buyer's problem: Maximizing payoff

- The expected payoff at t of a buyer using $\sigma \in \Sigma$ is

$$u_t^{\mathcal{M}_p}(\theta_t;\sigma) = E\left[e^{-(r+\rho)(\tilde{\mu}_\sigma - t)}\left(\tilde{\theta}_{\tilde{\mu}_\sigma} - p(\tilde{\mu}_\sigma)\right) \mid \tilde{\theta}_t = \theta_t\right],$$

  where $\tilde{\mu}_\sigma$ is the stopping time determined by σ.
  - Discounting; Probability that the buyer survives in the market until $\tilde{\mu}_\sigma$
  - IC: $u_t^{\mathcal{M}_p}(\theta_t;x) = \nu_t^{\mathcal{M}_p}(\theta_t) := \sup_{\sigma\in\Sigma} u_t^{\mathcal{M}_p}(\theta_t;\sigma)$, for all t and θ_t

Seller's Problem

Seller's problem: Finding the profit-maximizing IC mechanism $\mathcal{M}_P$

- The expected profit at τ from a buyer arriving at τ:

$$\pi^{\mathcal{M}_p}(\tau) = E\left[e^{-(r+\rho)(\tilde{\mu}_x - t)} p(\tilde{\mu}_x)\right]$$

- The present value of total profit:

$$\Pi^{\mathcal{M}_p} = I\pi(0) + \int_0^\infty \lambda e^{-r\tau} \pi^{\mathcal{M}_p}(\tau) \mathrm{d}\tau.$$

Solution: Sales Policy

Solve the seller's problem using "allocation rule".

Given an IC mechanism, a sales policy is defined as

$$A = \{t \in \mathbb{R}_+ : x(\theta_L, t) = 1\}.$$

Denote a mechanism with a sales policy A by $\mathcal{M}_{P,A} = \langle p_A, x_A \rangle$.

- Consider sales policies that are closed sets WLOG.
- A can be described by a function $m_A : \mathbb{R}_+ \to \mathbb{R}_+ \cup \{+\infty\}$, such that the next date after t in A is

$$m_A(t) := \inf\{s \in A : s \geq t\}.$$

Optimal Prices for a Given Sales Policy: Overview

Derive optimal prices given a sales policy in two steps:

1. Deduce lower bounds for $\left(\nu_t^{\mathcal{M}_{P,A}}(\cdot)\right)_{t\geq 0}$, the value of a buyer's problem, in any mechanism implementing A.
2. Provide a mechanism where the buyer's payoffs achieve the bounds.

Lower Bounds for the Value of a Buyer's Problem

Lemma 1 (Lower Bounds for the Value of a Buyer's Problem)

The difference between expected rents for a high- versus low-value buyer at date t satisfies

$$\nu_t^{\mathcal{M}_{P,A}}(\theta_H) - \nu_t^{\mathcal{M}_{P,A}}(\theta_L) \geq e^{-(r+\rho+\alpha_L+\alpha_H)(m_A(t)-t)}(\theta_H - \theta_L).$$

Moreover, a low-value buyer's rent satisfies

$$\nu_t^{\mathcal{M}_{P,A}}(\theta_L) \geq \int_t^\infty \alpha_L e^{-(r+\rho+\alpha_L)(s-t)} \nu_s^{\mathcal{M}_{P,A}}(\theta_H) ds.$$

Remarks on the Lower Bounds

For the first inequality, consider a high type and a low type.

- Suppose the high type is not permitted to buy until either one of the two buyers' values changes, or until date $m_A(t)$.
- Conditional on values changing before $m_A(t)$, there is no difference in their rents (expected continuation payoffs at the first change).
- Conditional on no change before $m_A(t)$, the high type expects an additional rent of at least $e^{-(r+\rho)(m_A(t)-t)}(\theta_H - \theta_L)$, since she can purchase at $m_A(t)$. The prob. is $e^{-(\alpha_L+\alpha_H)(m_A(t)-t)}$.

For the second inequality, a low-value buyer can do no worse than to wait until his value turns high, and then follow an optimal strategy.

Optimal Prices for a Given Sales Policy

Lemma 2 (Optimal Prices for a Given Sales Policy)

Let A be any sales policy. Define, for all t,

$$p_A^*(t) = \theta_H - \int_t^\infty \alpha_L e^{-(r+\rho)(s-t)-(r+\rho+\alpha_L+\alpha_H)(m_A(s)-s)}(\theta_H - \theta_L)ds$$
$$- e^{-(r+\rho+\alpha_L+\alpha_H)(m_A(t)-t)}(\theta_H - \theta_L).$$

Let, for all t,

$$x_A^*(\theta_L, t) = \mathbf{1}\{t \in A\}, \quad x_A^*(\theta_H, t) = 1.$$

Then, $\mathcal{M}_{P,A}^* = \langle p_A^*, x_A^* \rangle$ maximizes the seller's expected profit conditional on implementing the sales policy A.

Separability of the Choice of the Sales Policy

A sequence of subproblems: At any sales date a, the seller must determine the next sales date.

Observation.
The choice of sales policy after a is separable from the choice before a.

- Suppose no subsequent sales. For low type at a, $\nu_a^{\mathcal{M}^*_{P,A}}(\theta_L) = 0$.
- Consider the effect of including sales on the rent to waiting buyers.
  - If $\nu_a^{\mathcal{M}^*_{P,A}}(\theta_L)$ increases by Δ, then the change in the expected payoff of a buyer arriving at $\tau < a$ is $\Delta e^{-(r+\rho)(a-\tau)}$.
- The effect on date-zero profits from all buyers arriving before a:

$$\Delta \left(I e^{-a(r+\rho)} + \int_0^a \lambda e^{-ra-\rho(a-\tau)}\, d\tau \right) = \Delta I e^{-ra}.$$

Proportional to I. Independent of the choice of sales dates before a.

Decomposition of the Value of Seller's Problem

Given a sale at a, the seller specifies the next sales date $a + z$. The value of the problem at a comprises:

1. Profits earned from buyers arriving b/t a and $a + z$ in case there are no sales after $a + z$ (so that $\nu_{a+z}^{\mathcal{M}^*_{P,A}}(\theta_L) = 0$)
2. Reduction in profits earned from buyers arriving before a due to including one more sale at $a + z$, rather than no additional sales
3. Discounted continuation value of the problem at $a + z$.

Backward Derivation

The profits from a buyer who arrives a length s before the next sale, conditional on the next sale being the last (using stationarity):

$$R(s) = \underbrace{\gamma^S \theta_H}_{\text{the surplus generated when the buyer's value is initially high}}$$

$$+ \underbrace{(1-\gamma^S)\left[\theta_H \int_0^s \alpha_L e^{-t(\alpha_L+r+\rho)} dt + \theta_L e^{-s(\alpha_L+r+\rho)}\right]}_{\text{the surplus generated when the buyer's value is initially low}}$$

$$- \underbrace{\gamma^S e^{-s(r+\rho)}(\theta_H - \theta_L)}_{\text{the buyer's discounted expected rent}} .$$

Effects of the Additional Sales

Suppose no further sales except sales at a and $a+z$.

- The date-a value of profits from buyers arriving b/t a and $a+z$:

$$\phi(z) = \int_0^z \lambda e^{-r\tau} R(z-\tau) d\tau.$$

- If exactly one sale after a, at $a+z$, the rent of a buyer with θ_L at a:

$$\nu_a^{\mathcal{M}^*_{P,A}}(\theta_L) = \Pr\left(\tilde{\theta}_{a+z} = \theta_H \mid \tilde{\theta}_a = \theta_L\right) e^{-z(r+\rho)}(\theta_H - \theta_L).$$

- Including the sale at $a+z$ reduces profits from waiting buyers by

$$I \Pr\left(\tilde{\theta}_{a+z} = \theta_H \mid \tilde{\theta}_a = \theta_L\right) e^{-z(r+\rho)}(\theta_H - \theta_L)$$

(effect measured in date-a terms).

Seller's Problem in Recursive Form

Seller's problem of choosing the next sale after a sales date a:

$$W^*(a) = \sup_{z>0} \left\{ \phi(z) - I \Pr\left(\tilde{\theta}_{a+z} = \theta_H \mid \tilde{\theta}_a = \theta_L\right) e^{-z(r+\rho)}(\theta_H - \theta_L) + e^{-rz} W^*(a+z) \right\}.$$

- Since $W^*(a)$ is independent of a, it is given by

$$W^* = \sup_{z>0} \left\{ \frac{\phi(z) - I \Pr\left(\tilde{\theta}_z = \theta_H \mid \tilde{\theta}_0 = \theta_L\right) e^{-z(r+\rho)}(\theta_H - \theta_L)}{1 - e^{-rz}} \right\}.$$

Solution to Seller's Problem

Proposition 1 (Optimal Sales Policy)

The optimal sales policy A^* is determined as follows.

Case (i): If $\frac{\theta_H}{\theta_L} \leq \underline{\theta}$, then $A^* = \mathbb{R}_+$; i.e., low-value buyers always buy.

Case (ii): If $\underline{\theta} < \frac{\theta_H}{\theta_L} < \overline{\theta}$, then there exists a unique scalar $z^* > 0$ such that $A^* = \{iz^* : i \in \mathbb{N}\}$; i.e., low-value buyers buy periodically.

Case (iii): If $\frac{\theta_H}{\theta_L} \geq \overline{\theta}$, then either $A^* = \varnothing$ or $A^* = \{0\}$; i.e., low-value buyers either never buy, or they buy only at date zero.

Corollaries

Corollary 1

The time between sales z^* in Case (ii) is increasing in $\dfrac{\theta_H}{\theta_L}$ over $(\underline{\theta}, \overline{\theta})$.

Corollary 2

Fix $\lambda, \alpha_L, r, \rho, \theta_H, \theta_L$ such that $1 < \frac{\theta_H}{\theta_L} < \frac{r+\rho+\alpha_L}{\alpha_L}$. Now, consider varying α_H, while letting $\gamma = \frac{\alpha_L}{\alpha_L+\alpha_H}$ (for stationarity). For any $\varepsilon > 0$, there exists $\overline{\alpha}_H$ such that, if $\alpha_H > \overline{\alpha}_H$, then Case (ii) applies and $z^* < \varepsilon$.

As α_H grows large, (i) the seller optimally holds sales at regular intervals, and (ii) sales can be sufficiently frequent.

Constant Values Benchmark

Suppose no value change ($\alpha_L = \alpha_H = 0$), and $\gamma \in (0, 1)$.

- The optimal price is constant, either θ_L or θ_H.
  - If the seller can instead condition her offer on arrival dates (not necessarily a price path), then he does no better than to offer the static monopoly price at that date. Buyers make immediate purchases.
  - But this policy does *not* depend on arrival dates. The seller can get the same outcome for all arrival dates by committing to a price path with the static monopoly price.
  - Buyers find it optimal to make immediate purchases, since their values do not change. A constant price is optimal among all possible mechanisms, not just price paths.
- This logic applies in this setting with random exits.
  - Price fluctuations are not driven by random exits.

Shutting Down Dynamic Arrivals

Suppose $\lambda = 0$. The seller optimally chooses at most one sales date.

- The profit from choosing a is $IR(a)$.
- $IR(a)$ obtains its max at 0, or approaches its sup as $a \to \infty$.
- Low-value buyers never wait.

If $a = 0$ maximizes $IR(a)$, then it goes to Case (iii).

- Deb (2014) studies a model where all buyers arrive at date zero, have a continuum of values, and where values change at most once.
- Deb suggests this as a theory of "introductory pricing".

Shutting Down Exogenous Exits

Exogenous exits are important for the stationarity.

- With no exits, the population of buyers grows without bound, and the seller's problem is no longer stationary.
- The working paper version (Garrett 2011) shows that the optimal sales policy is always bounded.
  - The optimal price eventually remains constant and equal to θ_H.
- Sales at later dates are increasingly costly in terms of rents, compared with the efficiency gains from selling to low types.
  - All earlier buyers have the option to wait, and are at no risk of exiting.
- However, price fluctuations still arise at early enough dates.

Stationarity is not essential.